

# JOYKIRAT SINGH

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## EDUCATION

### University of North Carolina at Chapel Hill

Ph.D. in Computer Science (Advised by Prof. [Mohit Bansal](#))

Chapel Hill, NC

2025 – Present

### Indraprastha Institute of Information Technology, Delhi

B.Tech in Computer Science and Design (Silver medalist); GPA: 9.24/10.0

New Delhi, India

2019 – 2023

## RESEARCH INTERESTS

My research focuses on building reasoning models and agents that can reason more efficiently and in an interpretable manner. Additionally, I am interested in text-based optimization for agents like self-evolving harnesses.

## EXPERIENCE

### MURGe-Lab Group

Research Assistant (Advisor: Mohit Bansal)

Chapel Hill, NC

Aug 2025 – Present

- Improved the reasoning abilities of language models while enhancing their efficiency through adaptive difficulty-based attention compression. (COLM 2026)
- Developed a method that decouples an LLM agent into a belief state model and a policy model to maintain a structured approximation of the belief distribution. (Preprint)

### Google Cloud

PhD Intern (Host: Sravan Bodapati)

Sunnyvale, CA

May 2026 – August 2026

- RL-finetuned Gemini-3.5-Flash to reduce P90 latency by 30% on an NL-to-SQL agentic task while preserving task accuracy.
- Designed a hypothesis-and-verification loop that detects early error signals using error taxonomy classification in coding agent trajectories, enabling mid-trajectory correction and improving end-to-end task success.
- Developed an inference-time co-evolution framework where an agent and a verifier iteratively improve each other from past rollouts alone, with no ground-truth supervision or model weight updates.

### Microsoft Research

Research Fellow (Principal Researcher: Akshay Nambi)

Bengaluru, India

Jul 2024 – May 2025

- Enhanced large language models' capabilities through agentic RL-based tool integration (NeurIPS FoRLM Workshop 2025).
- Investigated LLMs' capacity to detect and correct reasoning errors, revealing fundamental weaknesses in their step-by-step reasoning processes (ACL Main 2025).
- Developed an automated, feedback-driven framework for task-aware and efficient prompt optimization (ACL Finding 2025).
- Enhanced small language models' mathematical reasoning through dense-reward MCTS training and Direct Policy Optimization for step-wise improvement.

### Laboratory for Computational Social Systems (LCS2), IIT Delhi

Graduate Research Assistant under Prof Tanmoy Chakraborty and Prof Soumen Chakrabarti

New Delhi, India

Jun 2023 – Jun 2024

- Revealed internal mechanisms driving step-by-step reasoning through mechanistic interpretability. (TMLR 2024)
- Improved arithmetic reasoning via symbolic solvers, low-rank adaptation, and RL (AAAI 2024).
- Developed a data-efficient task adaptation method that aligns model behavior through targeted neural rotations (EMNLP Finding 2025).

### Expedia Group

Software Development Engineer, Payment's Module

New Delhi, India

Aug 2023 – Jun 2024

- Contributed to payment and conversation modules using Java/Kotlin, implementing CI pipelines, Datadog metrics, and LanguageIO-based translation for production systems.

### Google Summer of Code

Intern, MusicBlock

Remote

May - Aug 2021

- Contributed to MusicBlock 2.0 using React, TypeScript, and P5, implementing MVVM-based palette and artboard features and improving UX from version 1.0.

## PUBLICATIONS

Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty

**Joykirat Singh**, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, Hyunji Lee, Elias Stengel-Eskin, Mohit Bansal (preprint) [[PDF](#)]

**Think Right: Learning to Mitigate Under-Over Thinking via Adaptive, Attentive Compression**

**Joykirat Singh**, Justin Chih-Yao Chen, Archiki Prasad, Elias Stengel-Eskin, Akshay Nambi, Mohit Bansal (Accepted COLM) [[PDF](#)]

**Agentic Reasoning and Tool Integration for LLMs via Reinforcement Learning**

**Joykirat Singh**, Raghav Magazine, Yash Pandya, and Akshay Nambi (Neurips FoRLM Workshop 2025) [[PDF](#)]

**Exposing the Achilles' Heel: Evaluating LLMs Ability to Handle Mistakes in Mathematical Reasoning**

**Joykirat Singh**, Akshay Nambi, Vibhav Vineet (ACL Main 2025) [[PDF](#)]

**PROMPTWIZARD: Task-Aware Prompt optimization framework**

Eshaan Agarwal, **Joykirat Singh**, Vivek Dani, Raghav Magazine, T. Ganu, Akshay Nambi (ACL Finding 2025) [[PDF](#)]

**Data-scarce Behavior Editing of Language Models**

**Joykirat Singh\***, Subhabrata Dutta\*, T. Chakraborty - (EMNLP Findings 2025) [[PDF](#)]

**How to think step-by-step: A mechanistic understanding of chain-of-thought reasoning**

Subhabrata Dutta\*, **Joykirat Singh\***, S. Chakrabarti, T. Chakraborty - (TMLR 2024) [[PDF](#)]

**Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning**

Subhabrata Dutta\*, **Joykirat Singh\***, I. Pandey\*, S. Manchanda, S. Chakrabarti, T. Chakraborty (AAAI 2024) [[PDF](#)]

**EROS: Entity-Driven Controlled Policy Document Summarization**

**Joykirat Singh\***, Sehban Fazili\*, Rohan Jain, Md. Shad Akhtar - (LREC-COLING 2024) [[PDF](#)]

## HONOR & AWARDS

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- Awarded a silver medal for academic excellence at IIITD for achieving highest CGPA in computer science branch
- Awarded the Chanakya Fellowship by iHub Anubhuti IIITD (2022) to work on an independent research problem
- Presented the paper (top ~2%, Oral Presentation), Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning at AAAI'24 Conference
- Volunteer, AAAI'24 Conference - received a travel grant to attend the conference

## PROFESSIONAL SERVICES

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Conference Reviewer

- COLM 2026
- NEURIPS 2026
- ACL 2025, EMNLP 2025 (ACL Rolling Review)
- ICLR 2024-25
- AAAI 2026

## REFERENCES

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- [Mohit Bansal](#), John R. Louise S. Parker Professor of CS, UNC Chapel Hill.
- [Sravan Bodapati](#), Sr. Staff Agentic AI SWE, Google
- [Akshay Nambi](#), Principal Researcher, Microsoft Research India.
- [Tanmoy Chakraborty](#), Associate Professor, IIT Delhi, India.
- [Soumen Chakrabarti](#), Professor, IIT Bombay, India.